

phone to enable haptic feedback. You would need discrete devices for that.

THE PROBLEM OF THE METAVERSE

The simple answer is that the smart-phone form-factor is the principal limitation that's preventing it from being the be-all and end-all of experiencing the Metaverse. What you need to experience the Metaverse is a standalone HMD that is untethered i.e. wireless. It should be independent of smart-phones, laptops or even PCs. They need to have plenty of cameras on the HMD itself to sense the surroundings and provide real 6DOF (six degrees of freedom) to the user. Then they need to have displays that are capable of not only rendering visuals at high resolutions but also at high refresh rates to



ensure a seamless experience. Also, it goes without saying that an untethered haptic feedback mechanism would go a long way towards ensuring proper immersion. And lastly, these HMDs need a way to communicate to the world, the non-verbal cues that the human wearer is displaying. Most experts argue that more than 70 per cent of all communication is non-verbal. So it goes without saying that missing out on the non-verbal cues would be a significant oversight.

So what's the best HMD to experience the Metaverse? The answer isn't simple. Since the Metaverse services out there right now are very pedestrian, you could simply get any of the existing HMDs and get the full experience. Mind you, the "full experience" here is something gamers have been experiencing for the past two decades. And it's not like the current Metaverse services are going to mature rapidly any time soon. Computing hardware manufacturers who're building the powerful

parallel processing engines that can power the Metaverse have gone on the record to say that the Metaverse depicted in science fiction is decades away because the hardware needed to power it properly is yet to exist. So it's going to be a slow ride and we're going to see multiple hardware manufacturers coming out with their Metaverse gear and we'll see plenty of iterations over the years as the tech gets better.

There isn't a single HMD that can check all the boxes yet. Most HMDs have nearly got the displays figured out with some having sorted the 6DOF issues. The onboard processors are still only capable of rendering dated visuals which look great as long as you maintain a certain distance from whatever you're looking at.

Companies such as Google, Meta, HTC, Apple and others are still working on prototypes of the headsets most apt to experience the Metaverse. Google is working on Project Iris which is reportedly an untethered device with external cameras for creating mixed reality (MR) environments. It's expected to offload the computing onto Google's data centres and will serve as a conduit.

Meta (formerly Facebook) is working on Project Cambria which has a high-resolution colour passthrough i.e. mixed reality engine and incorporates eye-tracking as well as face-tracking to convey non-verbal communication. It will feature multi-element pancake lenses that will make the HMD more compact than what the company has ever put out before.

HTC, one of the pioneers in HMD technology, is working on a wrist-based tracker that allows you to accurately portray the position of your hands in the virtual world



Credit: Bradley Lynch

without having to rely on external cameras. HTC is yet to shed light on their next-gen HMD after the recently launched Vive Flow so we don't know what they have in store.

Apple has remained tight-lipped about their upcoming headset although, we might head more during WWDC in June.



Lastly, Ultrahaptics, the company that acquired Leap Motion has been showcasing haptic feedback technology that does not require physical contact. STRATOS Explore uses the erstwhile Leap Motion sensor to track your hands in 3D space and then uses an array of ultrasound emitters to provide haptic feedback up in the air.

THE PERFECT HMD

Does not exist. In fact, there isn't anything in existence that comes close to providing even a sub-standard experience in the Metaverse. By that, we mean an HMD that can provide immersive visuals, track the wearer and their arms and legs, capture the environment and mix that with the virtual world, and read non-verbal cues that can be digitised. Each of the above facets is being covered by different manufacturers and it will be quite a while before they all converge to provide a uniform experience.

Until then, it seems that Meta and Google have the lead with the former going all-in into the Metaverse. The Metaverse is currently in its infancy, and we're a long way from seeing a common standard of experience or even the hardware required to sustain a virtual world proportionate to the world. If you're in the market to experience any of the current Metaverse services, then a simple laptop or desktop PC will suffice. Any of the existing HMDs can render the current-gen Metaverse in all its blocky glory. **d**